



"Call mom" → "Calling Tom..."



4-yr-old: "Play Elsa" → "Playing Alexa radio"



British accent: "tomato" → [confused silence]

**Today: why this is one of the
hardest problems in AI.**

How Voice Assistants Hear You

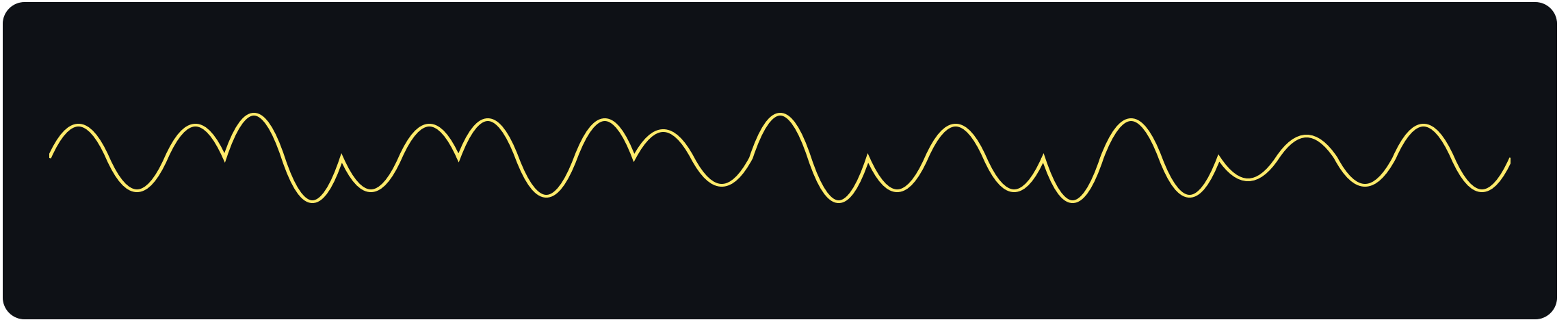
The hidden science of speech AI

DEEPAK PISKALA · 19 YEARS IN AI · AUTHOR, *BUILDING SPEECH AI*

THE QUESTION THIS BOOTH ANSWERS

**How does a microphone turn the words
"play despacito" into Despacito actually
playing?**

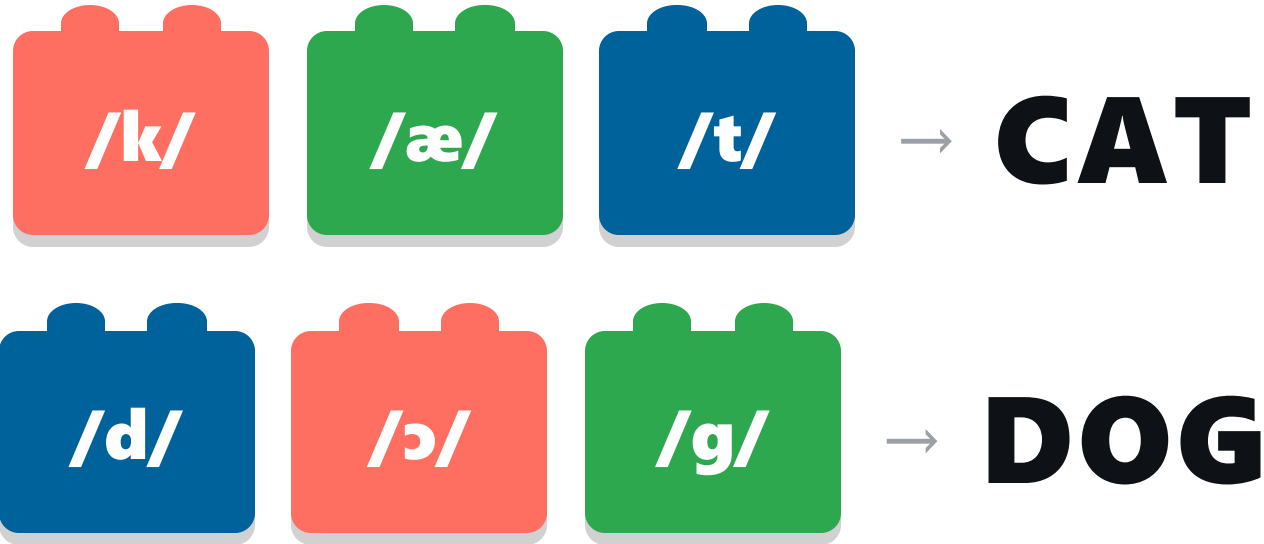
Your voice is just a wiggle.



A microphone records pressure changes ~16,000 times per second.

That's all your voice assistant gets. A wiggle.

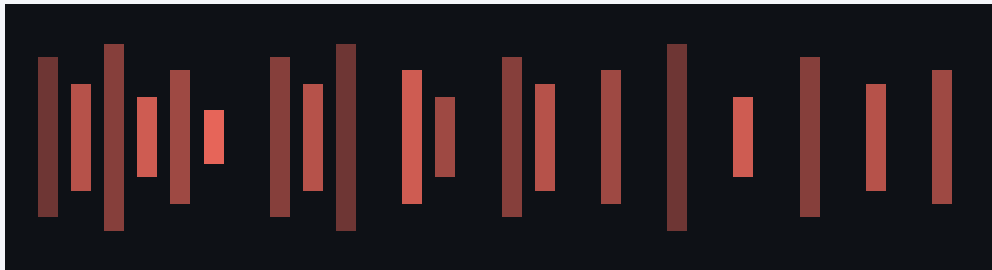
English has ~44 sounds. Every word is built from them.



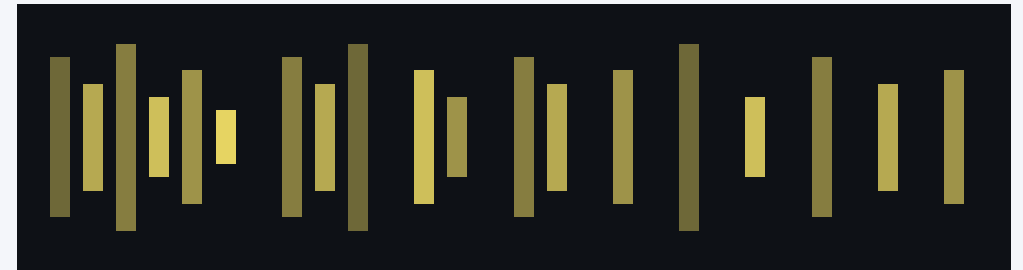
These tiny units are called **phonemes**. The first job of a voice assistant: **listen to the wiggle and guess which Legos.**

These two sentences sound **identical**.

"RECOGNIZE SPEECH"



"WRECK A NICE BEACH"



↑ Same waveform. **You** know which is which because you know the conversation. Machines are still catching up.

Inside Alexa, Siri, Google — the same 7-stage pipeline.

From the moment you speak the wake word to the moment the device speaks back, your voice passes through seven specialised AI systems — each one a research field of its own.



⚡ WHY THIS MATTERS

If *any one* of these 7 stages misfires, the whole experience breaks. ASR mishears "mom" as "Tom" → you call the wrong person. That's why voice AI is fragile even after decades of research.

🎓 EACH BOX = A STEM CAREER

Signal processing (wake word), deep learning (ASR), linguistics (NLU), reinforcement learning (dialog), software engineering (skills), generative AI (NLG), audio synthesis (TTS).

For 30+ years, voice AI was **many small models stitched together.**

RULES ERA

1950s – 1970s

"If it sounds like *this*, then it's *that*."

Engineers hand-built rules and sound templates. Brittle. Worked for a tiny vocabulary in a quiet room (Bell Labs' 1952 "Audrey" knew only spoken digits). Failed the moment you had a cold.

PIPELINE ML

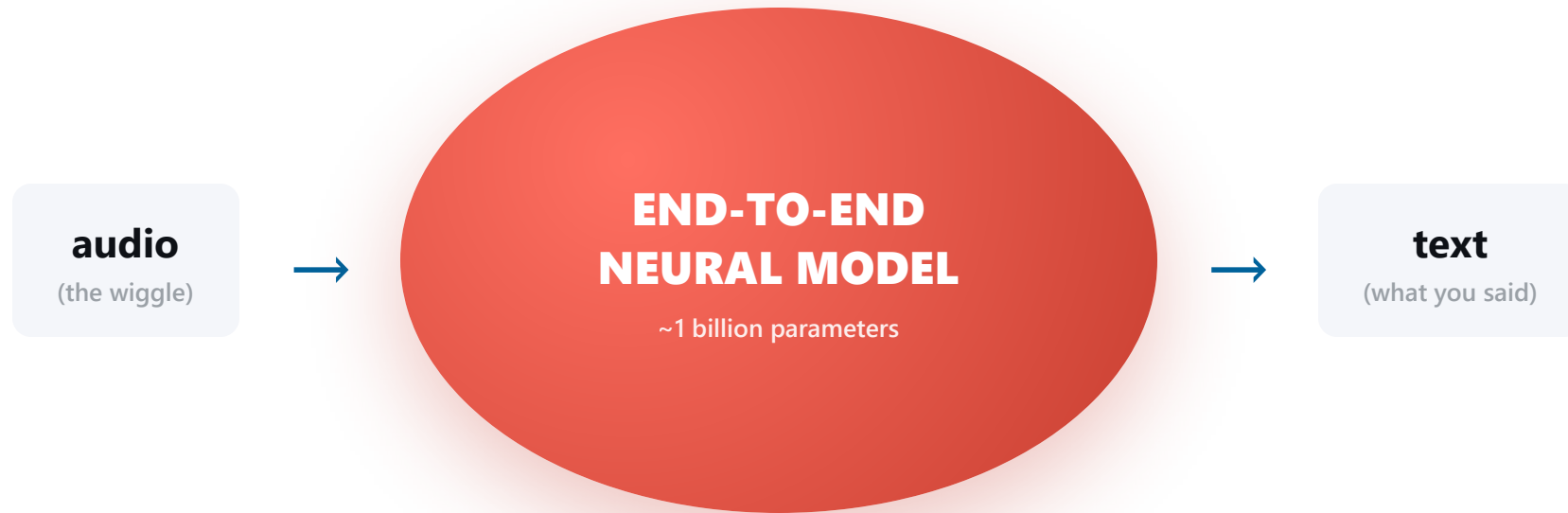
1980s – mid-2010s

A factory line of small ML brains, each doing one job.



Each box was its own model, trained separately. If one was weak, the whole pipeline got worse. Errors compounded.

Today: **one giant brain** learns the whole job.



No hand-written rules. No separate pipeline stages. **The model figures it out from millions of examples.** That shift, plus far more training data, is why today's recognizers handle accents, noise, and casual speech much better than a decade ago.

⚡ TRY IT YOURSELF — 3 MINUTES

Let's break it on purpose.

Open the voice-typing feature on a phone or Chromebook (or any free browser-based recognizer) and try to fool it three ways.



The Whisperer

Speak a sentence at half-volume. Watch which words go missing.



The Accent

Same sentence — different regional accent. Note which words slip.



The Noisy Room

Speak with background music. Watch the model hallucinate lyrics.

↳ Try it live with me in this booth: Fri 23 Oct, 12:45–1:30 PM ET

Voice assistants are **worse** at understanding the people in your room.

2-3x

**higher error rate
on children's speech
vs. adults.**



Shorter vocal tracts

Higher pitch, different formants
— most models weren't trained
on them.



Developing pronunciation

"R", "S", "Th" come online at
different ages.



Less training data

Privacy rules limit children's
audio in datasets.

Your students are the underserved users — and the future researchers who fix this.

Source: The Learning Agency, "Closing the Child Speech Recognition Gap": about 5% word error rate on adults vs. 11–18% on children for the same models.

The future isn't voice alone. It's **multimodal**.

When you talk to a friend, they don't *just* hear you. They see your face, follow your gaze, watch your hands, remember the conversation. The next generation of AI does the same.



Voice

What you said



Vision

Lip-reading, gaze, who's in the room



Gesture

Pointing, nodding, sign language



Context

What you said 5 minutes ago, where you are

▼ **fused inside one model** ▼

A truly conversational AI understands you the way **a person** would.

Six unsolved problems waiting for **your students**.



LINGUISTICS + ML

Low-resource languages

7,000+ human languages. Only a small fraction have enough data for voice AI to work well. Whole communities are locked out.



SOCIO-LINGUISTICS

Code-switching

Bilingual speakers mix languages mid-sentence (Spanglish, Hinglish). Models break. Humans don't.



AFFECTIVE COMPUTING

Emotion & intent

Sarcasm. Urgency. Distress. Same words, different meaning. Today's AI hears the words and misses the human.



DEVELOPMENTAL AI

Voices for everyone

Children, elderly, speech disorders. Build assistants that work for the people they currently fail.



PRIVACY + EDGE ML

On-device recognition

Run capable speech models on small, low-power chips so audio never leaves the device. Privacy by default.



MULTIMODAL FUSION

Voice + vision + gesture

Combine signals the way humans do. Still wide open for new ideas.



Every one of these is a PhD thesis, a startup, and a 30-year career — waiting for a student you teach next semester.

TAKE THIS BACK TO YOUR CLASSROOM — 15 MIN, NO EQUIPMENT

Activity: "Be the Recognizer"

1

Pair students up

One speaker, one "recognizer" (writes down what they hear).

2

Add noise

Round 1 in a quiet room. Round 2 with music playing while the speaker whispers.

3

Tally errors

Compare what was said to what was written. Calculate the class-wide "word error rate".

4

Discuss

What made it hard? Where would *context* have helped? Bridge directly to how AI faces the same problem.



FREE WORKSHEET + TEACHER GUIDE

Download the PDF from this booth's Resources (printable, no logins)

FOR YOU, FOR YOUR CLASSROOM — ALL FREE

Resources



Teachable Machine (Google)

Train a voice classifier in a browser in 10 minutes. No code.



Hugging Face Spaces

Hundreds of free speech-AI demos to try in a browser. Preview before class.



Building Speech AI

Free companion code at github.com/prdeepakbabu/building-speech-ai



IEEE TryEngineering

Free, ready-to-use engineering lesson plans for pre-university classrooms.

All four open in a web browser. None require an account to explore.

**The next breakthrough in voice AI
is sitting in your **classroom**.**

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